Chapter 5: Information Theory for Learning

Part III — Statistical Foundations of Deep Learning & Reinforcement Learning

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Abstract

The loss that trains almost every classifier and language model — cross-entropy — is an information-theoretic quantity, and reading it that way explains what training actually does. Entropy measures the uncertainty in a distribution; cross-entropy measures the cost of using the *wrong* distribution to encode the right one; and the two differ by exactly the KL divergence. Because the entropy of the truth is a constant, minimising cross-entropy is minimising KL to the truth — the model is being pulled onto the data distribution, nothing more. This chapter proves that decomposition, shows why KL is non-negative (so cross-entropy can never beat entropy), connects mutual information to what a feature tells you about a label, and ends at perplexity, which is just cross-entropy wearing different units.

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1 Entropy, cross-entropy, and KL

Definition 1.1 (Entropy, cross-entropy, KL)**.** For distributions p, q over the same outcomes,

$$H(p) = -\sum_x p(x)\log p(x), \quad H(p,q) = -\sum_x p(x)\log q(x), \quad \mathrm{KL}(p\|q) = \sum_x p(x)\log\frac{p(x)}{q(x)}.$$

$H(p)$ is the **entropy** (uncertainty in p), $H(p,q)$ the **cross-entropy** (cost of coding p with a code built for q), and KL the **divergence** from p to q.

The three are one identity apart, and that identity is why cross-entropy is the loss.

Theorem 1.1 (Cross-entropy is entropy plus divergence)**.** $H(p,q) = H(p) + \mathrm{KL}(p\|q)$.

Proof. Expand the KL: $\mathrm{KL}(p\|q) = \sum_x p(x)\log p(x) - \sum_x p(x)\log q(x) = -H(p) + H(p,q)$. Rearranging gives the identity. □

When a classifier minimises cross-entropy against the data, p is the (fixed) true label distribution and q the model. Since $H(p)$ does not depend on the model, *minimising* $H(p,q)$ *is minimising* $\mathrm{KL}(p\|q)$ — the model is pulled toward the truth, and the best it can ever do is $H(p)$, the entropy floor. The middle panel of the figure is this exact picture: cross-entropy as a function of the model q bottoms out at $q = p$, touching the horizontal entropy floor, with the KL as the gap above it.

Theorem 1.2 (Gibbs' inequality)**.** $\mathrm{KL}(p\|q) \geq 0$, with equality iff $p = q$.

Proof. Using $\log t \leq t - 1$ for $t > 0$ (with equality at $t = 1$),

$$-\mathrm{KL}(p\|q) = \sum_x p(x) \log \frac{q(x)}{p(x)} \leq \sum_x p(x) \left(\frac{q(x)}{p(x)} - 1 \right) = \sum_x q(x) - \sum_x p(x) = 0.$$

So $\mathrm{KL} \geq 0$; equality needs $q(x)/p(x) = 1$ everywhere, i.e. $p = q$. □

So cross-entropy can never be smaller than entropy: no model encodes the data better than the data's own distribution. One caution the definition makes plain: KL is *not symmetric*, $\mathrm{KL}(p\|q) \neq \mathrm{KL}(q\|p)$ in general — which is why the forward KL (used in maximum likelihood) and the reverse KL (used in some variational methods) produce different fits.

2 Mutual information

Definition 2.1 (Mutual information)**.** $I(X;Y) = H(X) - H(X \mid Y) = \sum_{x,y} p(x,y) \log \dfrac{p(x,y)}{p(x)p(y)}$.

Mutual information is how much knowing Y reduces the uncertainty in X — the entropy of X minus the entropy that remains once Y is known. It is $\mathrm{KL}\big(p(x,y) \,\|\, p(x)p(y)\big)$, the divergence between the joint and the product of the marginals, so by Gibbs' inequality $I(X;Y) \geq 0$, zero exactly when X and Y are independent. This is the information-theoretic meaning of a useful feature: a representation Y is informative about a label X to the extent that $I(X;Y)$ is large. (The **information bottleneck** proposes that good representations maximise $I(Y;\text{label})$ while minimising $I(Y;\text{input})$ — a clean idea, though whether it *explains* deep learning is still debated, and we state it as a lens, not a law.)

3 Perplexity

Language models report **perplexity**, and it is not a new quantity — it is the cross-entropy, exponentiated.

Definition 3.1 (Perplexity)**.** For a model with per-token cross-entropy H (in nats), the perplexity is $\mathrm{PPL} = e^H$.

Perplexity is the *effective vocabulary size* the model is choosing among: a model as uncertain as a uniform guess over V tokens has cross-entropy $\log V$ and perplexity V; a perfect model has cross-entropy 0 and perplexity 1. The right panel plots the exponential link, with a uniform model over ten thousand tokens at perplexity 10,000 and a cross-entropy of two nats at perplexity 7.4. Reporting perplexity rather than loss changes nothing mathematically — $e^{(\cdot)}$ is monotone, so minimising one minimises the other — but it puts the number on the interpretable scale of "how many equally likely choices."

Summary

Entropy $H(p)$ is uncertainty; cross-entropy $H(p,q)$ is the cost of coding p with q; they satisfy $H(p,q) = H(p) + \mathrm{KL}(p\|q)$, so minimising cross-entropy against fixed data minimises KL to the truth, floored at $H(p)$. Gibbs' inequality gives $\mathrm{KL} \geq 0$ (equality iff $p = q$), so no model beats the data's own entropy, and KL's asymmetry separates forward- from reverse-KL fits. Mutual information $I(X;Y) = H(X) - H(X \mid Y) = \mathrm{KL}(p(x,y)\|p(x)p(y)) \geq 0$ measures what a feature says about a label. Perplexity is e^H — cross-entropy on the scale of effective

vocabulary size.